

CONFIDENTIAL

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RCA OS Network Constitution

Network Governance, Policy Framework & Architecture Standards

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Section 1: Executive Summary

The **RCA OS Network Constitution** establishes the authoritative governance framework for all network infrastructure owned, operated, and maintained by RCA Operations Services. This document serves as the single source of truth for network policy, security standards, architectural decisions, and operational procedures governing the RCA OS network environment.

Scope. This constitution applies to all network devices (routers, switches, wireless access points, monitoring appliances), all users (administrators, end users, third-

party vendors), and all services (routing, switching, DHCP, NAT, SNMP, syslog) operating within or connecting to the RCA OS network infrastructure.

Objectives. The governance framework defined herein is designed to achieve four core objectives:

- **Security:** Enforce defense-in-depth through VLAN segmentation, hardened access controls, SNMPv3 encryption, and stateful perimeter ACLs.
- **Reliability:** Ensure continuous network availability through dynamic EIGRP routing, proactive monitoring via Prometheus and Grafana, and defined incident response procedures.
- **Compliance:** Align all network controls to recognized frameworks including NIST CSF, CIS Cisco IOS Benchmarks, and PCI-DSS where applicable.
- **Scalability:** Provide a modular, well-documented architecture that supports controlled growth without compromising security posture.

Authority. This constitution is ratified and enforced by the IT Director in coordination with the Security Officer and Network Administrator. All personnel with network access are bound by the policies defined herein. Violations are subject to the enforcement actions specified in Section 3.4.

⚠ **Important**

This document is classified **CONFIDENTIAL — Internal Use Only**. It contains sensitive infrastructure details including IP addressing, VLAN architecture, ACL logic, and credential references. Distribution outside authorized personnel is strictly prohibited.

Section 2: Network Governance Framework

2.1 Governance Authority & Roles

The following table defines the governance roles, their responsibilities, and the scope of authority each role holds within the RCA OS network environment.

Role	Responsibilities	Authority Scope
Network Administrator	Day-to-day operations, configuration changes, port provisioning, DHCP management, monitoring review, first-line troubleshooting	Standard and emergency changes; proposes Normal/Major changes for approval
Security Officer	Policy enforcement, security audit, incident response lead, compliance verification, vulnerability assessment	Approves Normal changes; co-approves Major changes; authority to suspend access
IT Director	Strategic decisions, budget allocation, vendor relationship management, architecture direction, policy ratification	Final approval authority on Major changes, budget, and vendor contracts
End Users	Comply with Acceptable Use Policy, report incidents and anomalies, complete security awareness training	Access limited to assigned VLAN; no administrative privileges
Third-Party Vendors	Scoped access for contracted services only, comply with NDA and AUP, coordinate with Network Administrator	Time-limited, supervised access; NDA required prior to any access grant

2.2 Document Control

Version History

Version	Date	Author	Description
1.0	May 1, 2026	Network Administration Division	Initial release — full governance framework, security policies, architecture standards, and operational procedures

Review & Amendment Process

- **Mandatory Review Cycle:** Annual review required. Next scheduled review: May 1, 2027.
- **Triggered Review:** Initiated upon any major infrastructure change (new device deployment, architecture redesign, security incident requiring policy update).
- **Amendment Process:** Changes proposed by the Network Administrator, reviewed by the Security Officer, and approved by the IT Director. All amendments recorded in the Version History table above.

Section 3: Acceptable Use Policy (AUP)

3.1 Authorized Use

The RCA OS network is provided for authorized business purposes. All users must adhere to the following terms:

- **Business use is primary.** All network resources on VLANs 10 (Data), 20 (Voice), 30 (Printers), 40 (Cameras), and 99 (Management) are designated exclusively for business operations.

- **Limited personal use** is permitted on VLAN 50 (WiFi) only, provided it does not interfere with business operations, consume excessive bandwidth, or violate any other provision of this policy.
- **All network activity is monitored and logged.** Syslog data is forwarded to a centralized Loki instance; SNMP metrics are collected by Prometheus. Users should have no expectation of privacy when using RCA OS network resources.

3.2 Prohibited Activities

The following activities are strictly prohibited on the RCA OS network:

- Unauthorized network scanning, penetration testing, or packet capture
- Connecting unauthorized devices, including personal switches, routers, access points, or DHCP servers
- Attempting to bypass VLAN segmentation or access restricted management networks (VLAN 99)
- Sharing credentials or using default, weak, or previously compromised passwords
- Running unauthorized services, including web servers, file sharing applications, or cryptocurrency mining software
- Tampering with or disabling port security, DHCP snooping, BPDU guard, or any other network security control
- Exfiltrating, copying, or transmitting confidential data outside authorized channels
- Using network resources for illegal activity of any kind

3.3 BYOD & Guest Access

Category	Policy
Personal Devices (BYOD)	Restricted to VLAN 50 (WiFi) only. No

Category	Policy
	personal devices may connect to Data (10), Voice (20), Printer (30), Camera (40), or Management (99) VLANs under any circumstances.
Guest Access	Requires prior approval from the Network Administrator. Access is time-limited, restricted to VLAN 50, and logged. Guest credentials expire automatically after the approved duration.
Vendor Access	Requires NDA execution and coordinated, supervised access. Vendor sessions are logged and reviewed post-engagement.

3.4 Enforcement & Violations

 **Warning — Enforcement Schedule**

Violations of this Acceptable Use Policy are subject to progressive disciplinary action as follows:

Offense	Action
First Offense	Written warning issued; mandatory security awareness review required within 5 business days
Second Offense	30-day network access suspension; reinstatement requires Security Officer approval
Third Offense	Permanent network access revocation; formal disciplinary action per organizational HR policy
Critical Violation	Immediate access suspension and formal investigation. Applies to data exfiltration, malicious activity, or any action posing imminent risk to the network. Law enforcement referral as warranted.

Section 4: Security Policies & Standards

4.1 Authentication & Access Control

All administrative access to RCA OS network devices is governed by the following hardened authentication and access control standards:

Control	Standard	Implementation
Admin Authentication	Local user database, privilege level 15	<code>username admin secret (Type 5 encrypted)</code>
SSH Version	SSHv2 only; Telnet disabled	<code>ip ssh version 2</code>
RSA Key Size	4096-bit minimum	<code>crypto key generate rsa modulus 4096</code>
SSH Ciphers	AES-256-CTR, AES-128-CTR only	No CBC modes permitted
SSH MACs	HMAC-SHA2-256, HMAC-SHA2-512	No SHA-1 or MD5
SSH Key Exchange	DH Group 14 (SHA-256), Group 16 (SHA-512)	No weak Diffie-Hellman groups
Max SSH Sessions	3 concurrent unauthenticated	<code>ip ssh maxstartups 3</code>
Auth Retries	2 per session	<code>ip ssh authentication-retries 2</code>
Brute-Force Protection	Block 300s after 3 failures in 120s	<code>login block-for 300 attempts 3 within 120</code>
Session Timeout	10 minutes idle	<code>exec-timeout 10 0</code>
VTY Access	Restricted to VLAN 99 + transit /30	<code>ACL MGMT_ACCESS / MGMT_SSH</code>
Console Security	Local auth, no outbound transport	<code>transport output none</code>
Unused VTY Lines	Lines 5-15 fully locked	<code>transport input none, 1-second timeout</code>

4.2 Network Segmentation (VLAN Architecture)

The RCA OS network employs strict VLAN-based microsegmentation to isolate traffic by function and enforce least-privilege access. The complete VLAN architecture is defined below:

VLAN	Name	Subnet	Purpose	DHCP	Security Controls
10	DATA	192.168.10.0/24	User workstations	Dynamic (.10-.254)	Port security max 3, voice VLAN overlay
20	VOICE	192.168.20.0/24	VoIP phones	Dynamic (.10-.254)	Auto QoS VoIP, TFTP option 150
30	PRINTERS	192.168.30.0/24	Network printers	Static only	Port security max 1, shutdown violation
40	CAMERAS	192.168.40.0/24	IP security cameras	Dynamic (.10-.254)	ACL CAMERA_RESTRICT, port security max 1
50	WIFI	192.168.50.0/24	Wireless / BYOD	Dynamic (.10-.254)	AP trunk with native VLAN 999
99	MANAGEMENT	192.168.99.0/24	Switch, router, monitoring	Static only	SSH-only, ACL-restricted
999	BLACKHOLE	None	Unused ports / native VLAN	None	Ports shut down, VLAN not routed

i Design Note

VLAN 1 is never used. All unused ports are assigned to VLAN 999 (BLACKHOLE) and administratively shut down. The native VLAN on all trunk links is set to 999 to mitigate VLAN hopping attacks.

4.3 SNMP Security

Simple Network Management Protocol access is tightly controlled. Legacy SNMP versions are explicitly disabled to prevent cleartext community string exposure.

Parameter	Configuration
Protocol Version	SNMPv3 only — v1/v2c explicitly disabled, all community strings removed
Security Level	authPriv — authentication and encryption are mandatory
Auth Protocol	SHA
Privacy Protocol	AES-128
Access Control List	ACL <code>SNMP_ACCESS</code> — permits only 192.168.99.10 (monitoring server)
Username	<code>rcamonitor</code> (not visible in <code>show running-config</code> ; verify with <code>show snmp user</code>)

4.4 Port Security Standards

Port Type	Max MACs	Violation Mode	Additional Controls
User Desk (Gi1/0/1-5)	3	Restrict	PortFast, BPDU Guard, aging 60s
Printers (Gi1/0/6-7)	1	Shutdown	PortFast, BPDU Guard
Cameras (Gi1/0/8-12)	1	Shutdown	PortFast, BPDU Guard, <code>CAMERA_RESTRICT</code> ACL
WiFi AP (Gi1/0/13)	N/A (trunk)	N/A	Allowed VLANs 50, 99 only; native 999
Monitoring (Gi1/0/14)	1	Shutdown	PortFast, BPDU Guard
Unused (Gi1/0/15-47)	N/A	N/A	Shut down, VLAN 999, <code>nonegotiate</code>

4.5 Layer 2 Security

The following Layer 2 hardening controls are implemented across the RCA OS switching infrastructure to mitigate common L2 attack vectors:

- **DHCP Snooping:** Enabled on VLANs 10, 20, 40, and 50. Trusted port: Gi1/0/48 (uplink to router). All other ports are untrusted by default, preventing rogue DHCP server attacks.
- **BPDU Guard:** Enabled on all edge (access) ports. Prevents rogue switch or access point injection by shutting down any port that receives a BPDU.
- **Spanning Tree Protocol:** Rapid PVST+ with root bridge priority 4096, ensuring RCA-SW01 is the deterministic root bridge for all VLANs.
- **VTP Mode:** Transparent. Prevents VLAN database propagation attacks and ensures the switch does not accept unsolicited VLAN changes from other devices.

4.6 Logging & Audit Trail

Control	Configuration
Config Archive	All configuration changes archived with hidekeys (archive log config)
Login Logging	Success and failure events logged to syslog
Auth Failure Rate Limit	5 per minute maximum
Syslog Destination	Centralized Loki instance at 192.168.99.10:1514/UDP
Log Retention	90 days in Loki; 64 KB rolling buffer on each device
Severity Level	Informational (level 6)
Timestamps	Service timestamps enabled (datetime msec); NTP-synchronized

Section 5: Regulatory Compliance Alignment

5.1 NIST Cybersecurity Framework (CSF) Mapping

The following table maps RCA OS network controls to the five functions of the NIST Cybersecurity Framework, demonstrating alignment with federal cybersecurity best practices:

NIST Function	Category	RCA Implementation
IDENTIFY (ID)	Asset Management	VLAN segmentation, port descriptions, SNMP sysName/sysLocation on all devices
PROTECT (PR)	Access Control	SSH hardening, SNMPv3 authPriv, MGMT ACLs, port security, brute-force lockout
PROTECT (PR)	Data Security	AES-128 SNMP encryption, AES-CTR SSH ciphers, MD5 EIGRP authentication
DETECT (DE)	Continuous Monitoring	Prometheus SNMP polling, Loki syslog aggregation, Grafana dashboards
DETECT (DE)	Anomaly Detection	Interface error rate alerts, EIGRP neighbor flap detection, login failure alerts
RESPOND (RS)	Incident Response	Port-security shutdown violations, brute-force lockout, WAN inbound ACL deny logging
RECOVER (RC)	Recovery Planning	Config archive, NTP sync for forensic timestamps, documented rollback procedures

5.2 CIS Cisco IOS Benchmark Compliance

The RCA OS network implements the following controls from the Center for Internet Security (CIS) Cisco IOS Benchmark:

CIS ID	Control	Status
1.1	Enable secret set (Type 5 encryption)	✓ Compliant
1.2	No plaintext passwords in running config (service password-encryption)	✓ Compliant
2.1	SSH version 2 enforced	✓ Compliant
2.2	Telnet disabled on all VTY lines	✓ Compliant
2.3	HTTP/HTTPS server disabled	✓ Compliant
2.4	CDP disabled globally	✓ Compliant
2.5	IP source-route disabled	✓ Compliant
2.6	BOOTP server disabled	✓ Compliant
3.1	Logging to external syslog server	✓ Compliant
3.2	Service timestamps enabled	✓ Compliant
3.3	Login failure logging	✓ Compliant
4.1	NTP configured with Google public servers	✓ Compliant
5.1	SNMP community strings removed (v3 only)	✓ Compliant
5.2	SNMP ACL restricts to monitoring host	✓ Compliant

5.3 PCI-DSS Relevant Controls

Where payment card data may traverse the RCA OS network, the following PCI-DSS requirements are addressed by the current architecture:

PCI Requirement	Description	RCA Implementation
Req 1	Install and maintain a	WAN_INBOUND ACL,

PCI Requirement	Description	RCA Implementation
	firewall configuration	NAT/PAT, VLAN segmentation
Req 2	Do not use vendor-supplied defaults	All defaults changed; VLAN 1 disabled; default passwords replaced
Req 6	Develop and maintain secure systems	SSH hardened; all unused services disabled (HTTP, HTTPS, CDP, BOOTP)
Req 8	Identify and authenticate access	Unique admin accounts; brute-force protection; session timeout
Req 10	Track and monitor all access	Syslog to Loki; config archive; NTP-synchronized timestamps
Req 11	Regularly test security systems	Prometheus/Grafana continuous monitoring; anomaly detection

Section 6: Edge Configuration Architecture

6.1 Edge Design Philosophy

The edge device **RCA-RTR01** (Cisco C1111X, IOS-XE) serves as the sole perimeter device between the internal RCA OS network and the untrusted WAN (Xfinity gateway at 10.0.0.1). It enforces defense-in-depth at the network boundary through Network Address Translation (NAT), stateful Access Control Lists (ACLs), and controlled route advertisement via EIGRP. All internet-bound traffic from every internal VLAN must traverse this device.

6.2 WAN Connectivity

Parameter	Configuration
Upstream Gateway	Xfinity gateway at 10.0.0.1 (DHCP reservation for 10.0.0.2 recommended; bridge mode preferred to avoid double NAT)
WAN Interface	GigabitEthernet0/0/0
NAT Type	PAT overload using WAN interface IP
NAT Scope	All internal VLANs: 192.168.10-50.0/24 + 192.168.99.0/24

6.3 WAN Security Controls

Control	Implementation	Purpose
Inbound ACL (WAN_INBOUND)	Applied to Gi0/0/0 inward	Defense-in-depth perimeter filtering
RFC 1918 Anti-Spoofing	Deny 172.16.0.0/12, 192.168.0.0/16 from WAN	Prevent spoofed source address attacks
Stateful Return Traffic	permit tcp any any established	Allow return traffic for established sessions only
ICMP Filtering	Allow echo-reply, unreachable, time-exceeded only	Enable diagnostics without exposure
Deny-All Default	deny ip any any log	Log all unauthorized inbound access attempts
CDP Disabled	no cdp enable on WAN interface; no cdp run globally	Prevent topology information leakage

6.4 Edge Routing (EIGRP)

- **EIGRP AS 100** configured on the transit link only (Gi0/1/0 → switch Gi1/0/48)
- **Default route:** 0.0.0.0/0 via 10.0.0.1, redistributed into EIGRP with seed metric
- **Passive-interface default** — only the transit link is an active EIGRP participant
- **MD5 authentication** on the EIGRP adjacency (key chain `RCA_EIGRP_KEY`)

- **EIGRP event logging:** Neighbor changes and warnings logged to syslog
- **SNMP traps:** EIGRP traps enabled for proactive monitoring

6.5 Edge Port Map

Interface	Role	IP Address	Security Controls
Gi0/0/0	WAN uplink	10.0.0.2/24	NAT outside, WAN_INBOUND ACL, no CDP
Gi0/1/0	Transit to switch	192.168.1.1/30	NAT inside, EIGRP MD5 auth
Gi0/1/1-7	Unused	None	Administratively shutdown

6.6 Edge Monitoring Integration

- **SNMPv3 authPriv** polled by Prometheus via SNMP Exporter every 60 seconds
- **Syslog** forwarded to Promtail at 192.168.99.10:1514 via Gi0/1/0 transit link
- **SNMP traps:** Link up/down, configuration changes, CPU threshold, memory threshold, syslog, EIGRP events

Section 7: Network Core Configuration Architecture

7.1 Core Design Philosophy

The core device **RCA-SW01** (Cisco Catalyst 3750X) operates as a collapsed core/distribution/access layer switch. It performs all inter-VLAN routing via Switched Virtual Interfaces (SVIs), serves as the DHCP server for all dynamically addressed VLANs, and enforces microsegmentation through port security, DHCP snooping,

BPDU guard, and inter-VLAN access control lists. All user-facing, device, and infrastructure ports terminate directly on this switch.

7.2 Layer 3 Routing

- **ip routing enabled** — full Layer 3 forwarding across all SVIs
- **EIGRP AS 100** advertises all 6 VLAN subnets plus the transit /30 to the edge router
- **Default route (0.0.0.0/0)** learned dynamically from the router via EIGRP `redistribute static`
- **No static routes required** — fully dynamic routing topology
- **Passive-interface default** — only Gi1/0/48 (uplink) participates in EIGRP
- **MD5 authentication** matches the edge router key chain (RCA_EIGRP_KEY)

7.3 DHCP Services

Pool Name	Network	Gateway	DNS Servers	Lease	Option 150
DATA_POOL	192.168.10.0/24	.1	8.8.8.8, 8.8.4.4	8 days	No
VOICE_POOL	192.168.20.0/24	.1	8.8.8.8, 8.8.4.4	8 days	Yes (TFTP for phone provisioning)
CAMERA_POOL	192.168.40.0/24	.1	8.8.8.8, 8.8.4.4	8 days	No
WIFI_POOL	192.168.50.0/24	.1	8.8.8.8, 8.8.4.4	4 days	No

i Note

Printers (VLAN 30) and Management (VLAN 99) use static IP assignments exclusively — no DHCP pool is configured for these VLANs. Refer to Appendix A for the static IP assignment registry.

7.4 Core Port Map

The following table documents every port on RCA-SW01, its assigned role, VLAN membership, switchport mode, and security controls:

Ports	Role	VLAN(s)	Mode	Security Controls
Gi1/0/1-5	User desks (PC + VoIP phone)	10 (data) + 20 (voice)	Access + Voice	Port-sec max 3, restrict, auto QoS VoIP, PortFast, BPDU Guard
Gi1/0/6-7	Printers	30	Access	Port-sec max 1, shutdown, PortFast, BPDU Guard
Gi1/0/8-12	IP Cameras	40	Access	Port-sec max 1, shutdown, CAMERA_RESTRICT ACL, PortFast, BPDU Guard
Gi1/0/13	WiFi AP	50, 99 (trunk)	Trunk	Allowed VLANs 50+99, native 999, PortFast trunk, BPDU Guard
Gi1/0/14	Monitoring server	99	Access	Port-sec max 1, shutdown, PortFast, BPDU Guard
Gi1/0/15-47	Unused — secured	999	Access	Shut down, nonegotiate, blackhole VLAN
Gi1/0/48	Uplink to router	Routed /30	No switchport	EIGRP MD5 auth, DHCP snooping trust

7.5 Inter-VLAN Access Control

The `CAMERA_RESTRICT` ACL is applied inbound on the Vlan40 SVI to enforce camera network isolation:

Rule	Action	Description
Permit DNS	Allow UDP/TCP 53	Camera name resolution
Permit NTP	Allow UDP 123	Camera time synchronization
Permit Intra-VLAN	Allow 192.168.40.0/24 to 192.168.40.0/24	Camera-to-camera communication
Permit to VLAN 99	Allow 192.168.40.0/24 to 192.168.99.0/24	Access to management/NVR
Deny to VLANs 10, 20, 30, 50	Deny + log	Block all inter-VLAN access to user/voice/printer/WiFi networks
Permit Any	Allow remaining	Permits internet-bound traffic via EIGRP default route

All other inter-VLAN traffic is permitted by default. Additional ACLs may be implemented as the security posture matures.

7.6 Core Monitoring Integration

- **SNMPv3 authPriv** polled by Prometheus via SNMP Exporter every 60 seconds
 - **Syslog** forwarded to Promtail at 192.168.99.10:1514, sourced from the Vlan99 SVI
 - **SNMP traps:** Link up/down, configuration changes, CPU threshold, VLAN create/delete, STP topology inconsistency, syslog events, EIGRP events
 - **Grafana dashboard:** Port status table, interface error rates, EIGRP neighbor and packet metrics, live syslog streams
-

Section 8: Monitoring & Observability Architecture

8.1 Monitoring Stack Components

The RCA OS monitoring infrastructure runs as a containerized stack on the monitoring server at 192.168.99.10 (VLAN 99, Gi1/0/14):

Component	Role	Container Name	Port
Prometheus	Metrics collection & time-series storage	rca-prometheus	9090
SNMP Exporter	Translates SNMPv3 polls to Prometheus metrics	rca-snmp-exporter	9116
Loki	Log aggregation & search	rca-loki	3100
Promtail	Syslog receiver → Loki shipper	rca-promtail	1514/UDP
Grafana	Dashboards & visualization	rca-grafana	3000

8.2 Data Flow

The monitoring architecture operates two distinct data pipelines:

Metrics Pipeline:

Cisco Devices → SNMPv3 Polls (60s interval) → SNMP Exporter (:9116) → Prometheus (:9090) → Grafana (:3000)

Logging Pipeline:

Cisco Devices → Syslog UDP 1514 → Promtail (:1514) → Loki (:3100) → Grafana (:3000)

i Architecture Note

Both pipelines converge in Grafana, enabling correlated views of metrics and logs on a single dashboard. Operators can investigate a CPU spike in Prometheus data alongside the corresponding syslog entries in Loki within the same time window.

8.3 Grafana Dashboard Panels (RCA Network Overview)

The primary Grafana dashboard contains 20+ panels organized into five logical rows:

Dashboard Row	Panels	Data Source
Device Health (6 panels)	Router uptime, Switch uptime, Router CPU %, Switch CPU %, Router memory %, Switch memory %	Prometheus
EIGRP AS 100 (8 panels)	Neighbor count (router), Neighbor count (switch), Retransmissions, Queued packets, Hold time remaining, Neighbor flaps from Loki, Packet rate chart (router), Packet rate chart (switch)	Prometheus + Loki
Interface Traffic (2 panels)	WAN interface traffic (Gi0/0/0 in/out), Transit link traffic (Gi0/1/0 ↔ Gi1/0/48 in/out)	Prometheus
Errors & Port Status (2 panels)	Interface error rates (CRC, input/output errors, drops), Port status table (all 48 ports, up/down/admin-down)	Prometheus
Syslog (5 panels)	Syslog ingestion rate by host, Volume by severity level, EIGRP event stream, Link/security event stream, Live syslog stream (all devices)	Loki

8.4 Data Retention

Storage Layer	Retention Period	Notes
Prometheus TSDB	90 days	Time-series metrics; older data automatically expired
Loki Logs	90 days	Compactor retention enabled; automatic cleanup
Device Log Buffer	64 KB rolling	Circular buffer on each Cisco device; oldest entries overwritten

Section 9: Change Management Policy

9.1 Change Categories

Category	Examples	Approval Required	Maintenance Window
Standard	Port description change, DHCP exclusion addition	Network Admin only	Business hours OK
Normal	VLAN addition, ACL modification, EIGRP tuning	Network Admin + Security Officer	After hours preferred
Emergency	Security incident response, critical outage fix	Network Admin (notify Security Officer within 1 hour)	Immediate
Major	Firmware upgrade, new device deployment, architecture change	IT Director + Security Officer + Network Admin	Scheduled maintenance window

9.2 Change Procedure

All changes to the RCA OS network must follow this standardized procedure:

1. **Document** the change request: what is being changed, why, when, and the rollback plan.
2. **Obtain approvals** per the category matrix defined in Section 9.1.
3. **Back up** the current `running-config` and `startup-config` on affected devices.
4. **Implement** the change during the approved maintenance window.
5. **Verify** the change using appropriate `show` commands and monitoring dashboards.
6. **Archive** the change via config archive (automatic — `archive log config` is enabled on all devices).
7. **Document** the outcome, any deviations from the plan, and close the change ticket.

9.3 Rollback Standards

⚠ **Mandatory Requirement**

Every change, regardless of category, **must** have a documented rollback procedure before implementation begins. No change may proceed without a viable rollback plan.

- **Pre-change backup:** Running-config backup is required before any configuration change.
- **Rollback timeline:** If rollback is needed, it must be completed within 15 minutes. If rollback cannot be completed within this window, escalate immediately per the incident response escalation path (Section 10.3).
- **Verification:** After rollback, verify the restored state via `show running-config`, EIGRP neighbor status, and Grafana dashboards.

Section 10: Incident Response Procedures

10.1 Severity Classification

Severity	Definition	Response Time	Examples
P1 — Critical	Complete network outage or active security breach	15 minutes	WAN down, EIGRP adjacency loss, unauthorized access detected
P2 — High	Major service degradation affecting multiple users or VLANs	30 minutes	VLAN unreachable, DHCP failure, sustained high CPU/memory
P3 — Medium	Minor service impact, single device or port affected	4 hours	Single port down, port-security violation, single camera offline
P4 — Low	Informational or cosmetic issue, no operational impact	Next business day	Port description mismatch, non-critical log anomaly

10.2 Response Workflow

8. **DETECT** — Grafana alert fires or user/stakeholder reports an issue, triggering investigation.
9. **TRIAGE** — Classify severity per the matrix above. Identify affected VLANs, devices, and services.
10. **CONTAIN** — Isolate the affected network segment (shutdown port, adjust ACL, disable interface) to prevent propagation.
11. **INVESTIGATE** — Review Loki syslog entries, Prometheus metrics, and relevant `show` commands to identify root cause.

- 12. **REMEDiate** — Apply the fix. Verify resolution via monitoring dashboards and show command validation.
- 13. **DOCUMENT** — Record incident details, root cause analysis, resolution steps, and lessons learned.
- 14. **REVIEW** — Post-incident review required for all P1 and P2 incidents within 48 hours of resolution.

10.3 Escalation Path

Severity	Primary Responder	Escalation
P4 / P3	Network Administrator	No escalation required unless unresolved beyond SLA
P2	Network Administrator	Security Officer notified immediately; joint investigation
P1	Network Administrator	Security Officer + IT Director notified immediately; all-hands response

Section 11: Physical Security & Environmental

11.1 Equipment Location

- All network equipment (RCA-RTR01, RCA-SW01, monitoring server) must be housed in a **secured, access-controlled room** with restricted key card or combination lock entry.
- Physical access is limited to **authorized IT personnel only**. A visitor log must be maintained for any non-IT access.

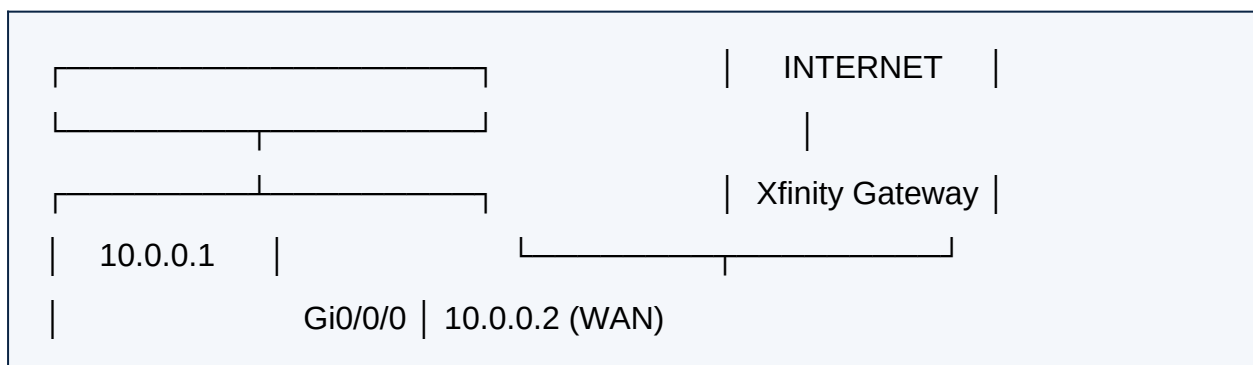
- The equipment room must have **adequate environmental controls**:
 - Active cooling (maintain 64–75°F / 18–24°C ambient temperature)
 - Uninterruptible Power Supply (UPS) with minimum 30 minutes runtime
 - Fire suppression system (clean agent preferred for electronics)

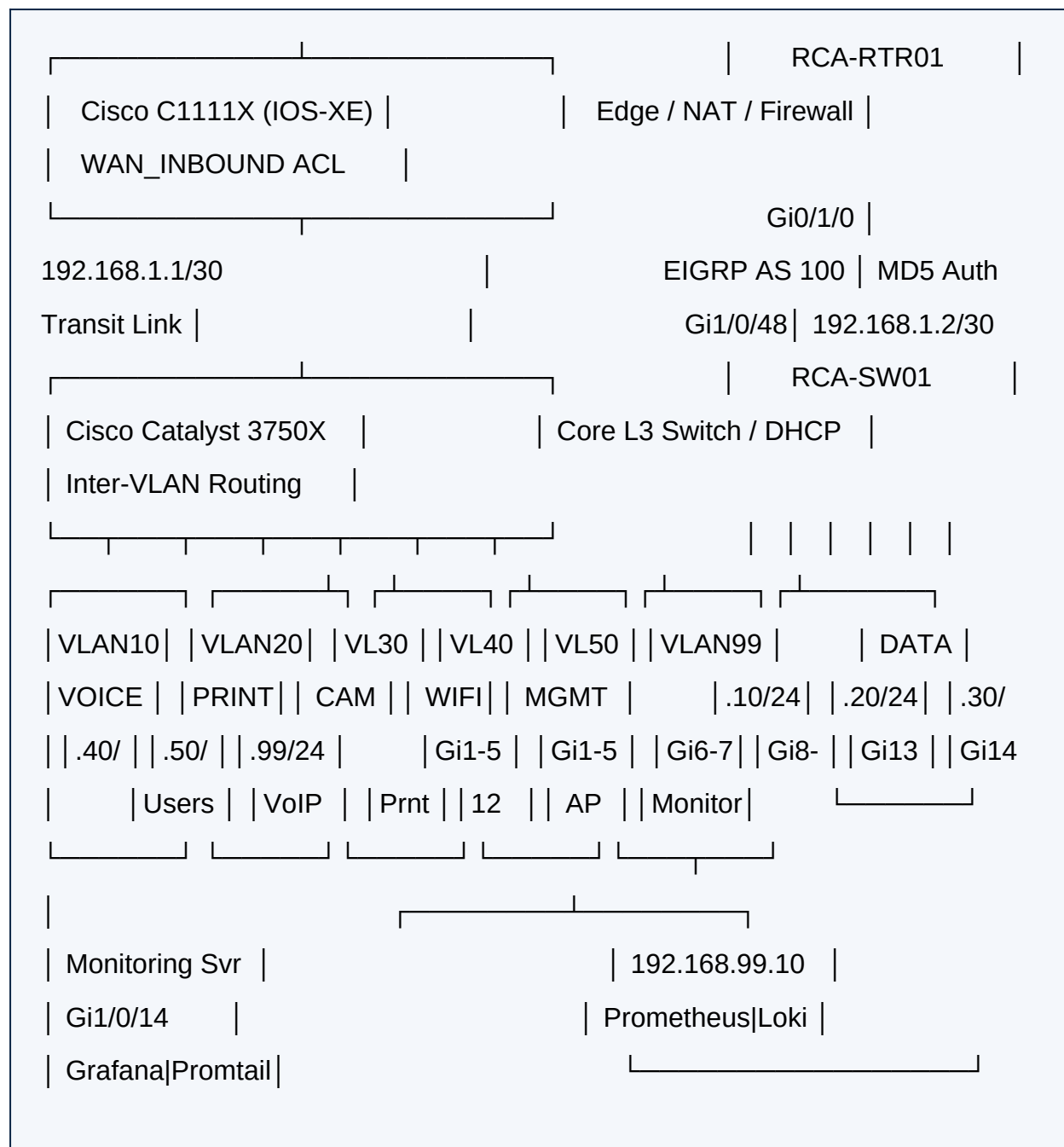
11.2 Cable Management

- All patch cables must be **labeled** with the port number and destination device at both ends.
- Unused ports should be **physically capped or covered** where possible to prevent unauthorized connections.
- Cable runs must be **documented** in the port map (Section 7.4) and kept current with any changes.
- Structured cabling standards (TIA-568) should be followed for all permanent installations.

Section 12: Network Topology Diagram Reference

The following diagram illustrates the physical and logical topology of the RCA OS network:





i Note

VLAN 999 (BLACKHOLE) is not depicted as it carries no traffic. All unused ports (Gi1/0/15-47) are assigned to VLAN 999 and administratively shut down. VLAN 1 is unused and has no SVI.

Section 13: Appendix

Appendix A: Static IP Assignment Registry

Device	IP Address	VLAN / Segment	Port	Notes
RCA-RTR01 WAN	10.0.0.2	N/A (WAN)	Gi0/0/0	DHCP reservation on Xfinity GW recommended
RCA-RTR01 LAN	192.168.1.1	Transit /30	Gi0/1/0	Point-to-point transit link to switch
RCA-SW01 Uplink	192.168.1.2	Transit /30	Gi1/0/48	Point-to-point transit link to router
RCA-SW01 Mgmt SVI	192.168.99.1	VLAN 99	SVI	Switch management IP address
Monitoring Server	192.168.99.10	VLAN 99	Gi1/0/14	Grafana / Prometheus / Loki / Promtail
Printer 1	192.168.30.10	VLAN 30	Gi1/0/6	Static IP configured on device
Printer 2	192.168.30.11	VLAN 30	Gi1/0/7	Static IP configured on device

Appendix B: Credential Reference

CRITICAL — Change Before Deployment

The credentials listed below are **placeholder values** used during initial configuration. Every credential in this table **MUST** be changed to a unique, strong value before the network is placed into production. Failure to do so

constitutes a critical security violation.

Credential	Default Placeholder Value	Used By
Enable Secret	RcaEn@ble2026!	Router + Switch (privilege exec)
Admin User	admin / RcaAdm1n2026!	Router + Switch (SSH login)
SNMPv3 Auth Password	Rca\$npAuth2026!	SNMP Exporter + Devices
SNMPv3 Priv Password	Rca\$npPriv2026!	SNMP Exporter + Devices
EIGRP Key String	RcaE1grp@uth2026!	Router + Switch (key chain)
Grafana Admin	admin / RcaGrafana2026!	Grafana web UI

Appendix C: Manual Commands Required

The following commands must be executed interactively in the CLI and cannot be pasted as part of a configuration block:

15. **RSA Key Generation** (both devices, exec mode):

```
crypto key generate rsa modulus 4096 label SSH-KEY
```

Note: This command is interactive and will prompt for confirmation. It must be run on both RCA-RTR01 and RCA-SW01.

16. **SNMPv3 User Creation** (both devices, config mode):

```
snmp-server user rcamonitor RCA_MON_GROUP v3 auth sha [AUTH_PASSWORD]  
priv aes 128 [PRIV_PASSWORD]
```

Note: This command does not appear in show running-config. Verify creation with show snmp user. Replace bracketed values with actual credentials.

Appendix D: Verification Commands Quick Reference

Command	Purpose
show ip eigrp neighbors	Verify EIGRP adjacency is established and stable
show ip route eigrp	Verify all EIGRP-learned routes are present
show snmp user	Verify SNMPv3 user was created successfully
show ip ssh	Verify SSH version 2 and configured algorithms
show port-security interface [intf]	Check port security status, MAC count, and violations
show ip dhcp binding	View active DHCP leases across all pools
show logging	Verify syslog configuration and recent log entries
show archive log config all	View complete configuration change history
show ip dhcp snooping	Verify DHCP snooping status and trusted ports
show spanning-tree summary	Verify root bridge status and STP mode
show vlan brief	Verify VLAN assignments across all ports
show ip nat translations	View active NAT translation table (router only)

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